# Vivek Damar

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## Education

**National Institute of Technology, Rourkela** — **Expected April 2027**
*Bachelor of Technology in Computer Science and Engineering* — *Rourkela, Odisha*

**Jawahar Navodaya Vidyalaya (JNV), Jhabua** — **May 2022**
*Class XII – CBSE, Science (PCM)* — *Jhabua, Madhya Pradesh*

**Jawahar Navodaya Vidyalaya (JNV), Jhabua** — **April 2019**
*Class X – CBSE* — *Jhabua, Madhya Pradesh*

## Relevant Coursework

- Database Management Systems
- Data Structures and Algorithms
- Operating Systems
- Object Oriented Programming
- Computer Networks
- System Design

## Work Experience

**Gazebo Labs Private Limited (IIT Bhubaneswar Research Park) – Certificate June 2026 – Aug 2026**
*Data Analyst Intern – Data & Business Intelligence Division* — *Bhubaneswar, India*

- Built Python and SQL **ETL pipelines** that ingested, cleaned, and validated high-frequency telemetry from smart IoT aquaculture devices – the curated layer every downstream dashboard read from.
- Added **data-quality validation** for missing, duplicate, and out-of-range sensor readings, stopping bad records before they reached reporting.
- Automated recurring reporting pipelines and built Python/SQL dashboards for real-time device-health visibility, cutting manual reporting turnaround by **70%**.
- Shaped raw telemetry into analysis-ready tables with **pandas** and SQL, and ran **exploratory data analysis** that surfaced operational trends and hardware faults early.

## Projects

**Distributed Job Scheduler** | *Node.js, TypeScript, PostgreSQL, Docker Compose* — **Live Demo** | **GitHub**

- Built a distributed cron-style **job scheduler and workflow orchestrator** on a persistent PostgreSQL queue – the same execution model batch data pipelines run on.
- Eliminated multi-worker race conditions with `SELECT ... FOR UPDATE SKIP LOCKED` and heartbeat-based lease renewal, ensuring each job was claimed by exactly one worker.
- Added exponential-backoff retries and a **Dead-Letter Queue** for failed jobs; verified fault tolerance under induced worker crashes via chaos testing.

**SeatWise** | *Next.js, PostgreSQL (Neon), Drizzle ORM, Stripe, Better-Auth* — **Live Demo** | **GitHub**

- Built a full-stack event ticketing platform with interactive seat maps, dynamic pricing, and real-time checkout.
- Eliminated double-booking under concurrent buyers using **atomic PostgreSQL transactions** (`SELECT ... FOR UPDATE`) for seat holds.
- Shipped Stripe Checkout with **idempotent** webhook handlers and server-verifiable QR check-in.

**FlagWise** | *TypeScript, Fastify, PostgreSQL, Redis, Docker* — **Live Demo** | **GitHub**

- Built a feature-flag microservice with REST APIs for flag config, targeting rules, and rollout percentages.
- Cut flag evaluation latency to under **5ms** with a Redis caching layer in front of the evaluation engine.
- Shipped production-ready with Docker Compose and **27+ Jest suites at 90%+ coverage**.

## Achievements/Certifications

**Problem Setter, The Timeless Saga** — Algorithmic & Programming Society, NIT Rourkela
Curated 30+ competitive programming challenges for the society's flagship contest.

**Hack Innovision Participant** — NIT Rourkela
Participated in the 36-hour annual national hackathon, prototyping full-stack software solutions. Certificate

## Technical Skills

**Languages**: Python, SQL, TypeScript, JavaScript (ES6+), C++

**Data & Analytics**: pandas, NumPy, ETL pipelines, data cleaning & validation, exploratory data analysis, SQL dashboards

**Databases**: PostgreSQL (transactions, row-level locking), Neon (serverless Postgres), Redis, Drizzle ORM

**Backend & APIs**: Node.js, Fastify, Express.js, RESTful APIs

**DevOps & Tooling**: Docker, Docker Compose, Git, Linux, Jest

**Frontend**: React.js, Next.js (App Router), Tailwind CSS